26 May 1970

MILITARY SPECIFICATION

LEATHER, CATTLEHIDE, CHROME TANNED, FATLIQUORED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the requirements for chrome tanned, full grain and burnished full grain cattlehide upper leather.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-L-43432 - Leather, Packaging and Packing Of.

STANDARD

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- The leather shall be full grain or burnished full grain cattlehide.

3.2 Tannage.- The leather shall be chrome tanned.

3.3 Physical requirements.-

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3.3.1 Trim (form) and area.- The leather shall be trimmed to conform with the definition of "side" as listed under "Definitions of Terms" in FED-STD-311. The area expressed in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather. The area of the side shall not be greater than 25 square feet.

3.3.2 Color.- The color of the leather shall be black. The leather shall be dyed on the grain side and the flesh side. The dye shall penetrate both sides of the leather, but need not be struck entirely through the leather. Leather previously dyed or finished in a color not specified and then redyed or refinished to match a specified color shall not be acceptable.

3.3.3 Finish.- The leather shall have a smooth bright pigmented finish on the grain side. Embossed or printed leather shall not be acceptable. A lacquer emulsion top finish based essentially on nitrocellulose compounding shall be applied to the grain side of the leather.

3.3.3.1 Burnishing.- When burnished leather is furnished, burnishing shall be accomplished with automatic machines equipped with 400 grit or 320 dull grit cloth or paper. The burnishing operation shall slightly level out the peak strata of the grain, but shall not remove all evidence of the grain. Snuffing or buffing to correct surface blemishes shall not be permitted.

3.3.4 Thickness.- The leather shall be 3-1/2 to 4-3/4 ounces in thickness. (NOTE: One ounce is equivalent to 1/64 inch.)

3.3.5 Stitch tearing strength.- When tested as specified in 4.3, not less than 80 percent of the specimens tested shall have minimum stitch tearing strength values of 30 pounds; the remaining 20 percent of the specimens tested shall have minimum stitch tearing strength values of 20 pounds.

3.3.6 Shrinkage temperature.- The leather shall have a minimum shrinkage temperature of 99°C when tested as specified in 4.3.

3.4 Chemical requirements.-

3.4.1 The leather shall conform to the chemical requirements of table I when tested as specified in 4.3.

TABLE I.- Chemical requirements

Characteristic	Minimum	Maximum
Chloroform - soluble material percent 1/	3.0	12.0
Chromic oxide, percent 1/	3.0	7.0
pH value	3.0	4.5

1/ Calculated on the moisture free basis.

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3.4.2 Paranitrophenol.- Unless otherwise specified (see 6.2), the leather shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.3.

3.5 Workmanship.- The leather shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with FED-STD-311, except when otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item.- The classification of defects found during the examination of the leather for visual and dimensional characteristics shall be in accordance with 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one side. The lot size shall be expressed in units of one side. Sampling for these examinations shall be performed in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 91 or more sides. Separate examinations shall be performed for visual and dimensional characteristics, but the same sample may be used for each examination.

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4.2.2.1 Examination for visual characteristics.-

Examine	Defect
Trim	Not in the form of sides.
Color	Not as specified or not uniform. Does not penetrate grain and flesh side. Evidence that leather has been redyed or refinished to cover unspecified color.
Finish	Finish not smooth. Dull finish. Finish peeled, chipped or tacky. Grain snuffed or buffed. Finish missing.
Quality of leather	Cut, hole, open or badly healed scratch, tear, stain, deep fat wrinkle, heavy vein, hip mark, rough shoulder, coarse grain, pipey area, loose flesh or other surface imperfection which individually or collectively impair the cutting value of more than 10 percent of the area of the leather.

4.2.2.2 Examination for dimensional characteristics.-

Examine	Defect
Area (expressed in terms of square feet)	Less than area marked on leather. Area exceeds 25 square feet.
Marking	Omitted, illegible or not in area specified.
Thickness 1/	Less than 3-1/2 ounces or greater than 4-3/4 ounces.

1/ Thickness shall be determined in accordance with method 1011 of FED-STD-311 except that no individual thickness measurement shall be less than the minimum or greater than the maximum specified.

4.2.3 Examination of preparation for delivery requirements.- An examination shall be made in accordance with the provisions of MIL-L-43432 to determine that packaging, packing and marking comply with the section 5 requirements of this specification.

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4.3 End item testing.- The methods of testing specified in FED-STD-311 wherever applicable and as listed in table II, shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exist:

a. Any test failure occurs for stitch tearing strength.

b. More than one test failure occurs for shrinkage temperature.

c. Any composite test result fails to meet the specified requirement.

TABLE II.- End item testing

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Composite
Material identification	3.1	1/	-	-
Tannage	3.2	1/	-	-
Finish	3.3.3	1/	-	-
Stitch tearing strength	3.3.5	2151	X	-
Shrinkage temperature	3.3.6	7011	X	-
Chloroform-soluble material	3.4.1	6341	-	X
Chromic oxide	3.4.1	6515	-	X
pH value	3.4.1	6621	-	X
Paranitrophenol	3.4.2	6711	-	X

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the requirement stated.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- Leather shall be packaged in accordance with the applicable requirements of MIL-L-43432.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C.- Leather shall be packed in accordance with the applicable requirements of MIL-L-43432.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-L-43432.

6. NOTES

6.1 Intended use.- The cattlehide is intended for use as upper leather in men's oxford safety shoes.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When paranitrophenol fungicide is not required (see 3.4.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Navy - SA, MC
Air Force - 82

Preparing activity:

Army - GL

Project No. 8330-0127

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and mail to preparing activity. Comments and information submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-L-43704 LEATHER, CATTLEHIDE, CHROME TANNED, PULLUP/RED

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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